

Repair Manuals

Timing chain: removal/installation

Warnings and recommendations

Note: Do not touch the terminals

An electrostatic discharge may occur

Before working on the fuel system, clean the area thoroughly

Note: Plug the ends of the fuel lines

Renew the seals

Always use new gaskets

Removal

Disconnect the battery earth cable

Place the bonnet in the service position

Remove the engine upper covers

Remove the strut brace connecting the shock absorbers

Remove the air filter housing

Remove the turbo inlet pipe

Remove the crankcase breather pipe from the cylinder head cover

Disconnect the engine wiring from the top of the engine

Remove the variable camshaft timing solenoids

Remove the ignition coils

Remove the spark plugs

Remove the fuel line between the high-pressure pump and the fuel rail

Note: Always renew the component

Remove the fuel rail

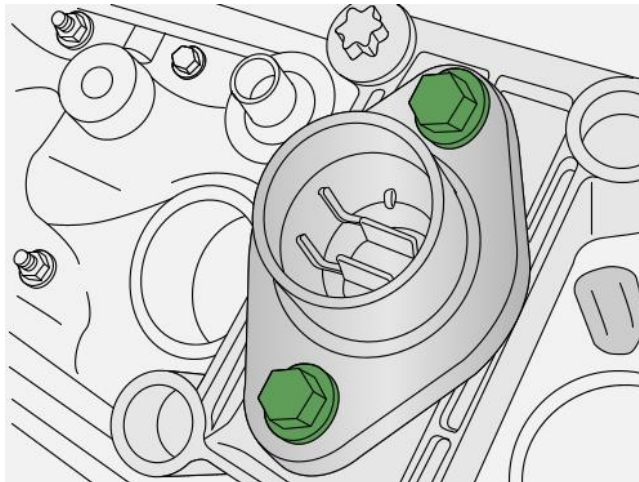
Remove the high-pressure pump

Remove the fuel feed line

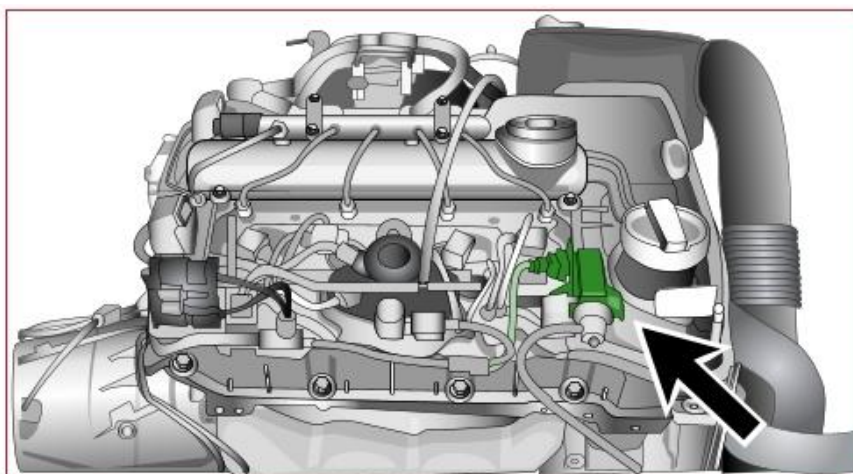
Remove the vacuum pump

Disconnect the oxygen sensor connector

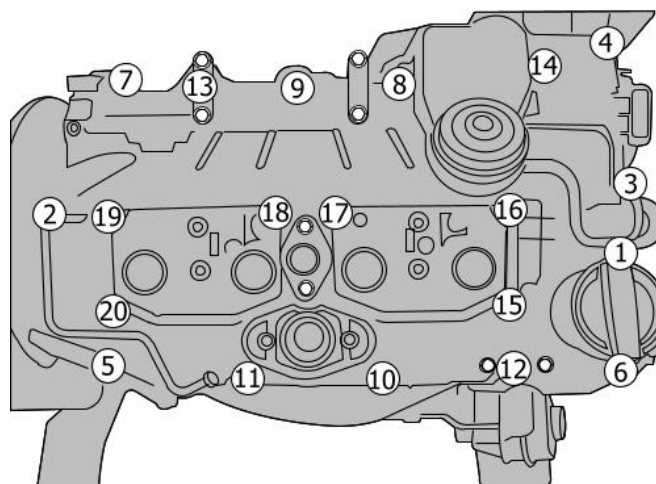
Remove the flange



Remove the pressure control valve

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Loosen the bolts in a criss-cross pattern, starting from the outside. Loosen approximately 1/2 a turn ((1 - 20)).

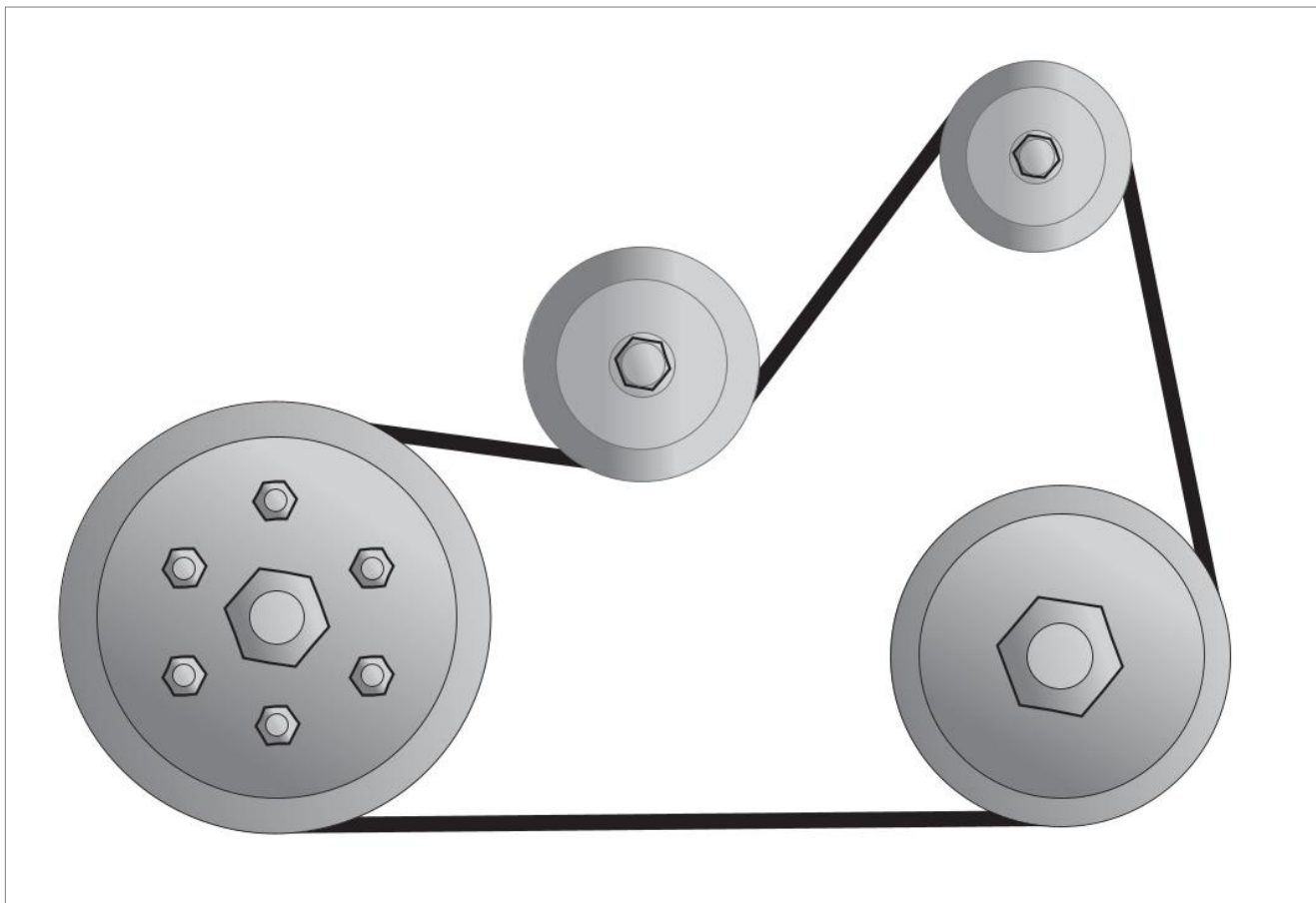


Remove the valve cover

Remove the ancillary drive belt

Drive belt layout

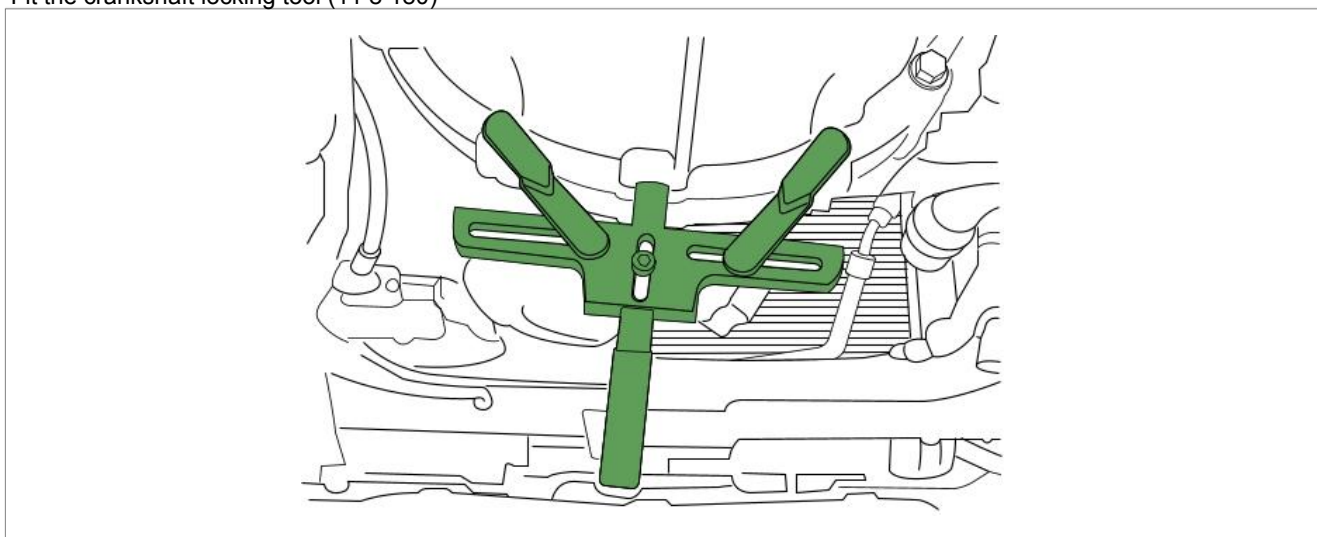
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Remove the engine lower covers

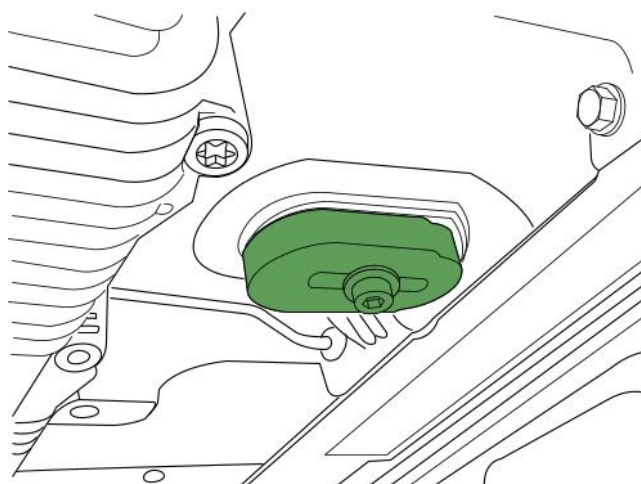
Vehicles with manual transmission:

Fit the crankshaft locking tool (11 8 180)



Vehicles with automatic transmission:

Fit the crankshaft locking tool (22 2 742)

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All vehicles:

Remove the crankshaft pulley outer bolts

Remove the crankshaft pulley

Remove the crankshaft locking tool

Fit the special tool to the crankshaft oil seal (50 Nm) (11 0 371, 11 0 372)

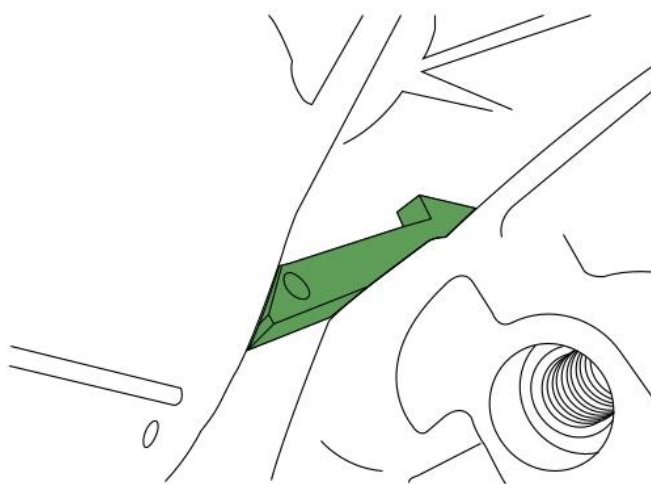
Remove and discard the crankshaft oil seal

Check the timing

Turn the camshaft to TDC for cylinder 1

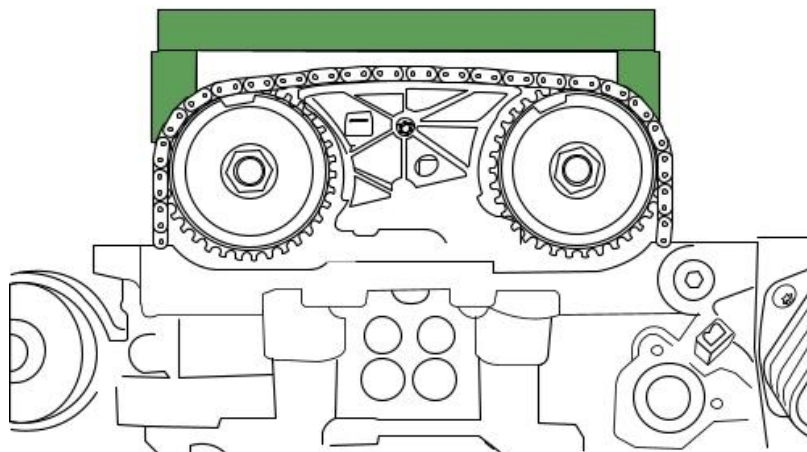
Remove the access plug

Fit the crankshaft locking tool (2 219 548)

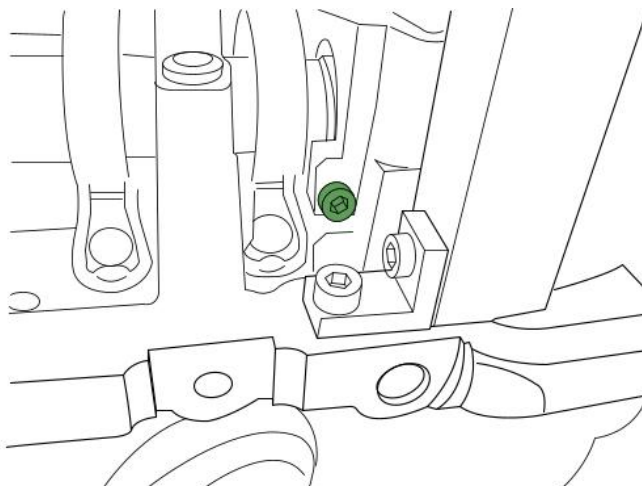


Fit the camshaft timing tool (2 212 831)

Note: Ensure correct fitting

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Secure the special tool with a bolt

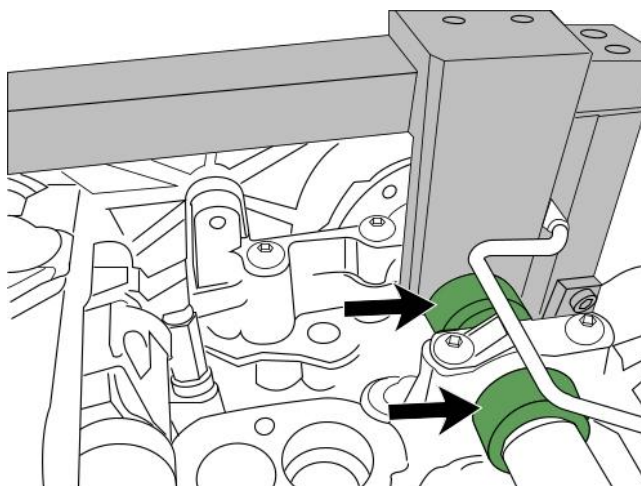


Lock the camshafts

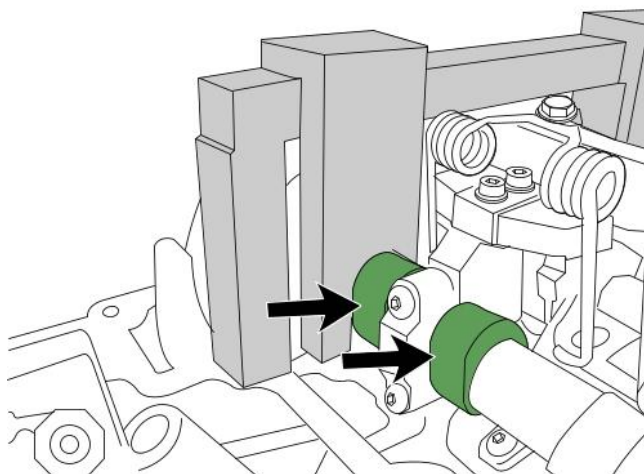
Check the position of the camshafts

Note: Cylinder 1 cams at TDC firing position

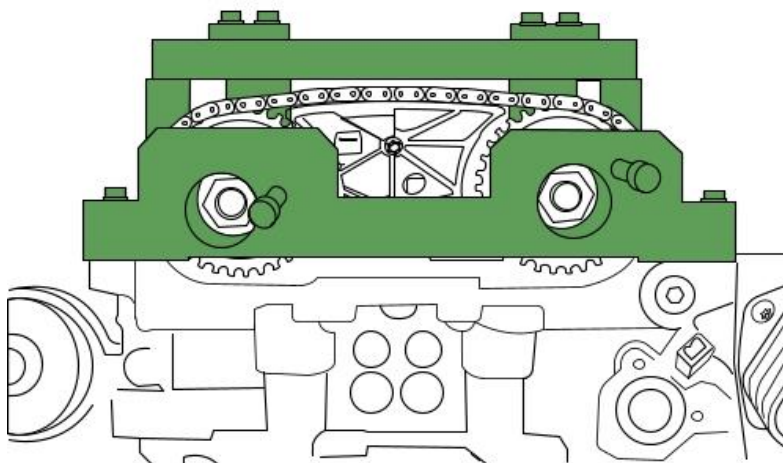
Exhaust camshaft:



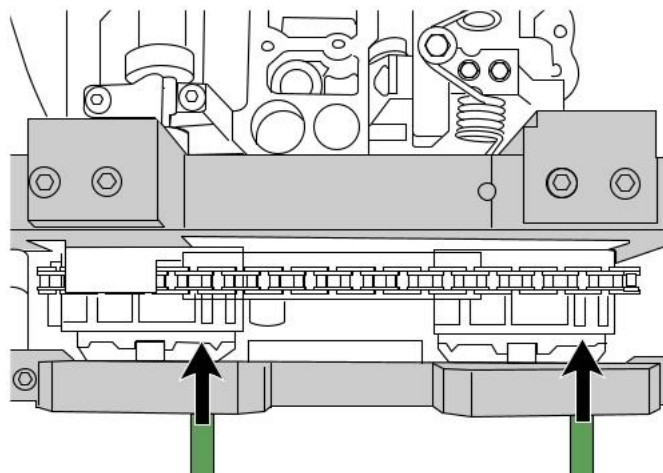
Inlet camshaft:

Repair Manuals

Fit the special tool (2 212 830)



Ensure that the locking tools fit correctly



Remove the special tools (2 212 831, 2 212 830)

Drain the engine oil

Convertible models:

Both sides:

Remove the diagonal struts

All vehicles:

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Support the engine

Disconnect the engine support

Lift the engine slightly (10 mm)

Remove the front wheels

Remove both front wheel arch liners

Vehicles with 4WD:

Drain the front differential oil

Remove the stabiliser links

Disconnect the track rod ends

Disconnect the lower suspension arms

Remove both front drive shafts

Remove the front propeller shaft

All vehicles:

Disconnect the steering column from the steering box

Disconnect the power steering wiring loom

Support the subframe (00 2 030, 31 5 250)

Remove the subframe bolts

Lower the front axle (< 100 mm)

Vehicles with 4WD:

Remove the front differential

Remove the drive shaft support

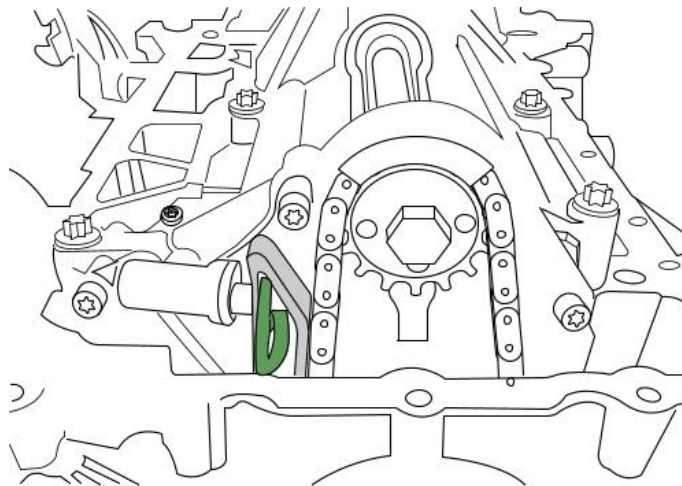
Note: Right-hand side

All vehicles:

Remove the oil sump

Compress the tensioner

Lock the tensioner; use a locking pin (11 4 120)



Vehicles with manual transmission:

Fit the crankshaft locking tool (11 8 180)

Vehicles with automatic transmission:

Fit the crankshaft locking tool (22 2 742)

Remove the crankshaft gearwheel centre bolt

Breakaway torque: (> 600 Nm)

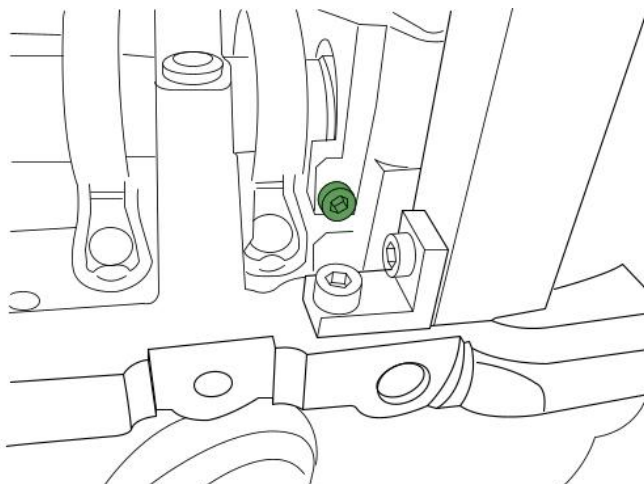
Remove the camshaft position sensors

Fit the camshaft locking tool (2 212 831)

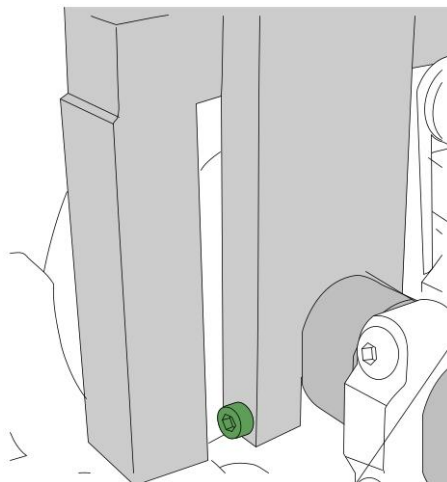
Check that the camshafts are locked

Exhaust camshaft:

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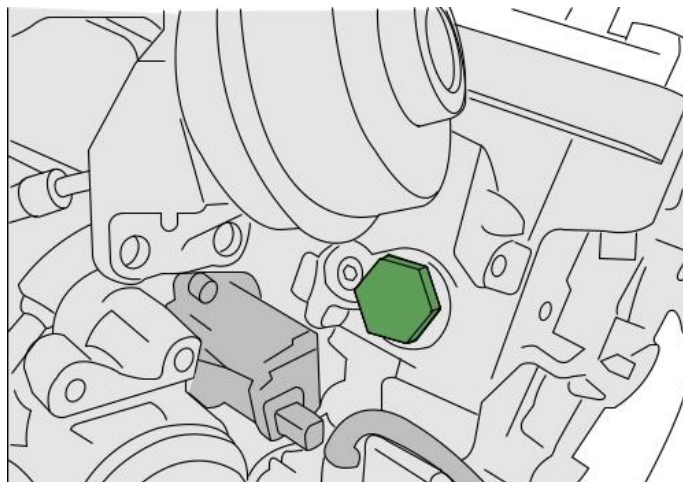
Inlet camshaft:



Remove the camshaft gearwheels

Disconnect the turbocharger actuator

Remove the chain tensioner

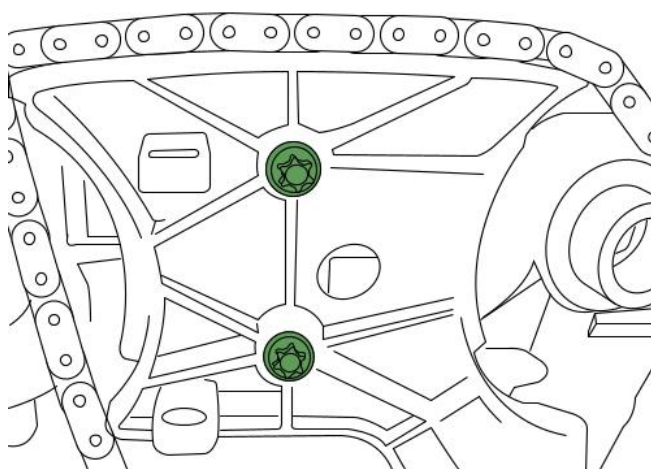
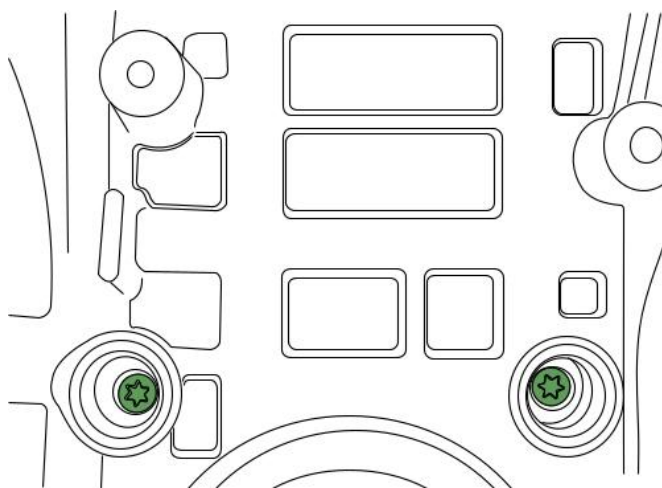
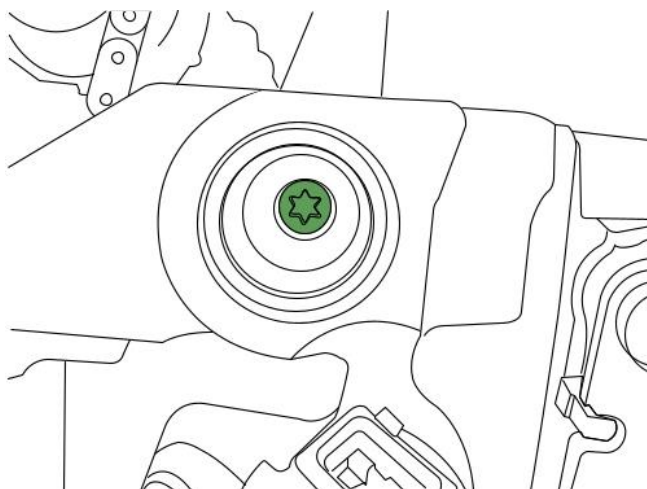


Remove the special tool (2 212 831)

Remove the crankshaft hub

Remove the access plugs

Remove the timing chain guide bolts from the cylinder head

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Remove the chain guide with the timing chain, guide rail and tensioner rail

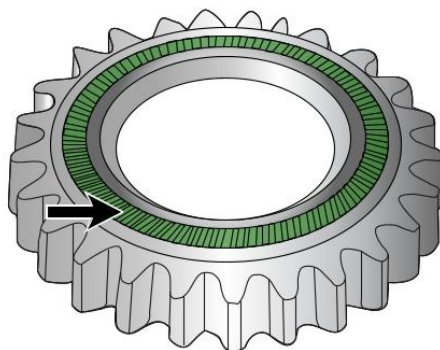
Installation

Timing chain:

Renew the crankshaft gearwheel

Note: Use an improved part

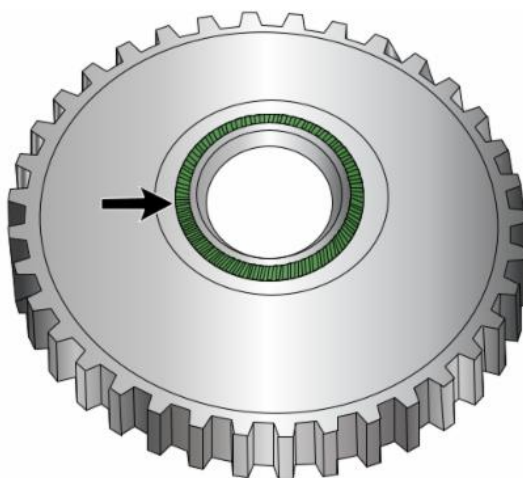
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Oil pump chain:

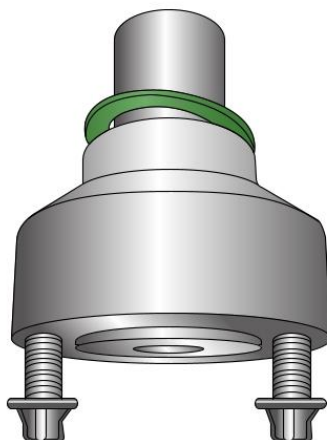
Check that the correct part has been fitted

Renew as necessary



Vehicles with manual transmission:

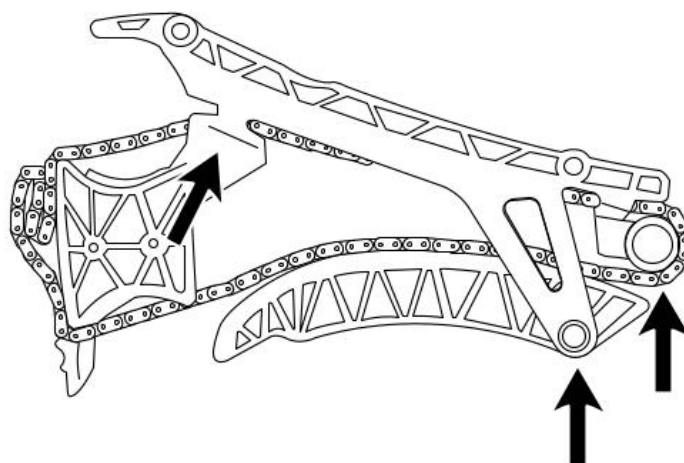
Remove the washer



All vehicles:

Fit the gearwheel together with the chain

Tension the chain

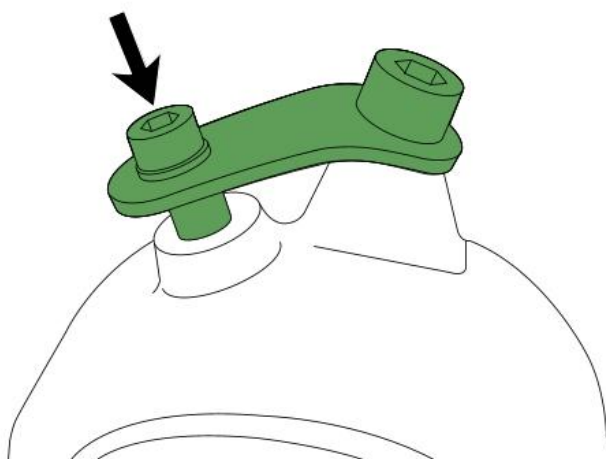
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Oil pump housing:

Remove the sealing cap

Fit the special tool (2 212 825)

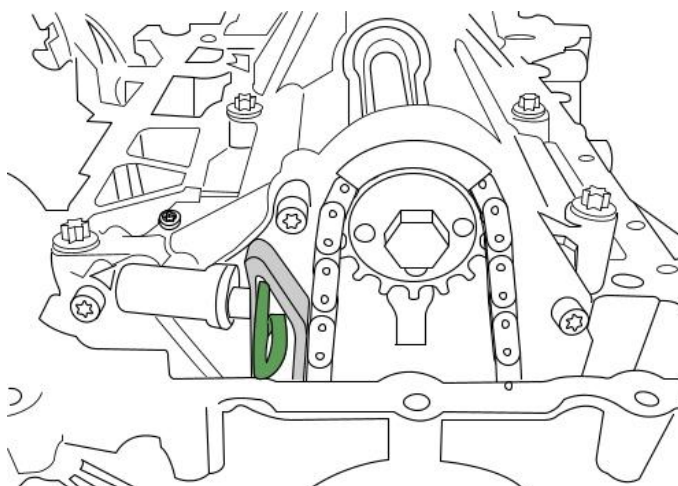
Turn the crankshaft until the locking pin fits



Refit the crankshaft hub

Fit the centre bolt of the crankshaft gearwheel/vibration damper

Remove the tensioner locking pin



Tighten the crankshaft pulley centre bolt

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Refit the camshaft gearwheels

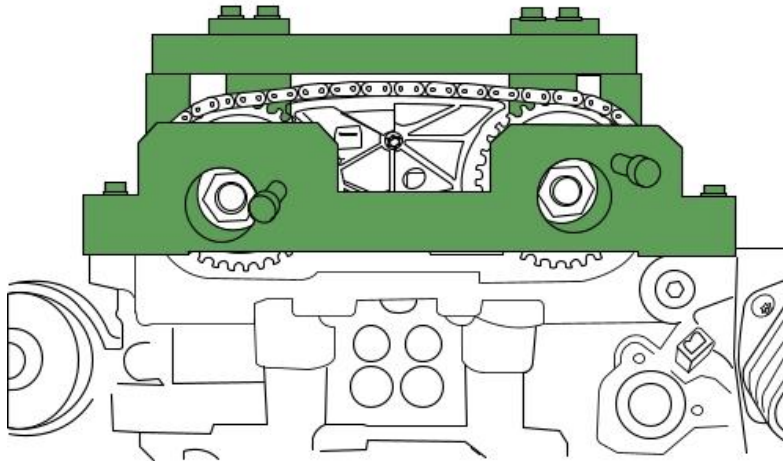
Check the marks

Exhaust camshaft gearwheel: ((EX))

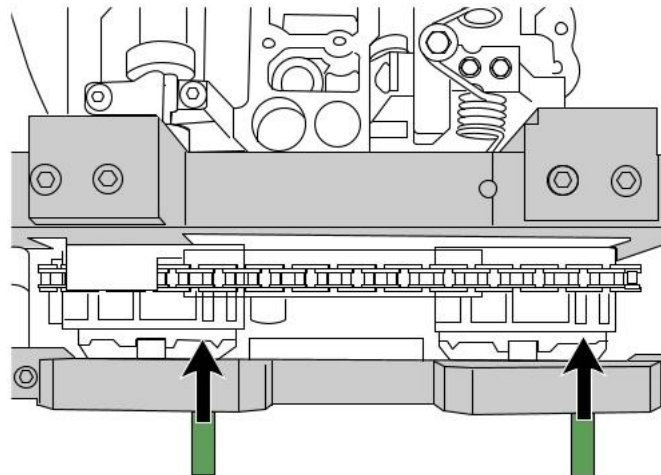
Inlet camshaft gearwheel: ((IN))

Note: Ensure correct fitting

Fit the special tools (2 212 831, 2 212 830)



Ensure that the locking tools fit correctly



Refit the camshaft gearwheel bolts

Note: Do not tighten the bolts

Fit the dummy chain tensioner (11 9 340)

Pre-tension the timing chain (0.6 Nm)

Use a special tool (00 9 250)

Tighten the camshaft gearwheel bolts

Remove the special tools (11 9 340, 2 219 548, 2 212 831, 2 212 830)

Compress the chain tensioner piston twice to drain the oil

Refit the chain tensioner

Turn the engine 2 rotations clockwise

Check the timing

If the locking tools do not fit, adjust the valve timing:

The crankshaft should be at TDC for cylinder 1

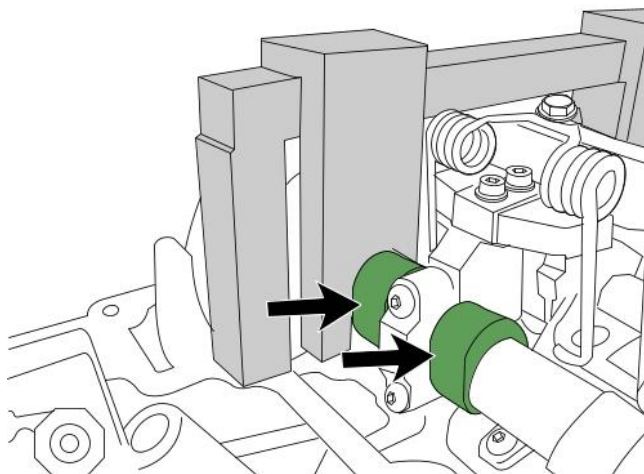
Ensure the crankshaft is locked (2 219 548)

Check the position of the camshafts

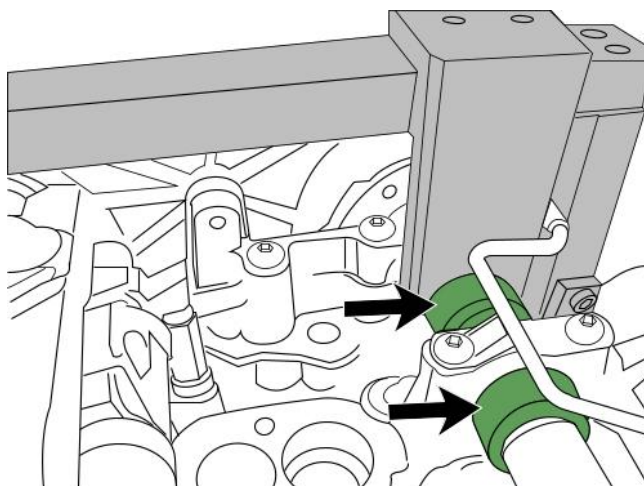
Inlet camshaft:

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Note: The cams for cylinder 1 should point straight up

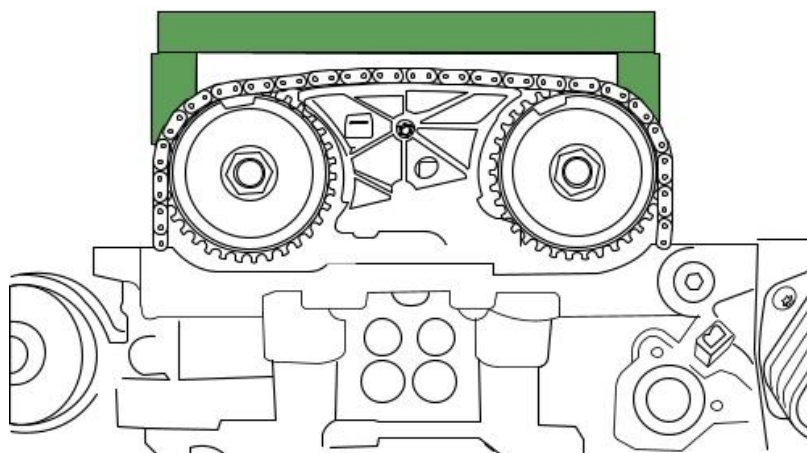


Exhaust camshaft:



Fit the camshaft timing tool (2 212 831)

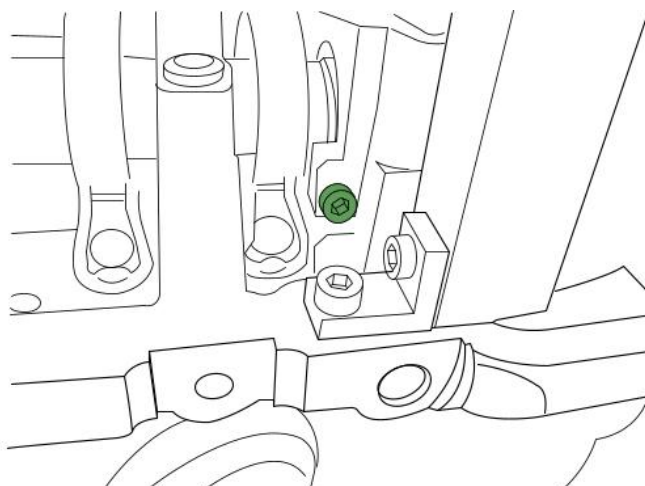
Note: Ensure correct fitting



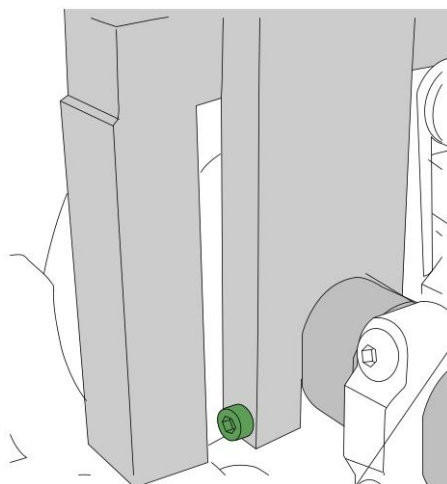
Check that the camshafts are locked

Exhaust camshaft:

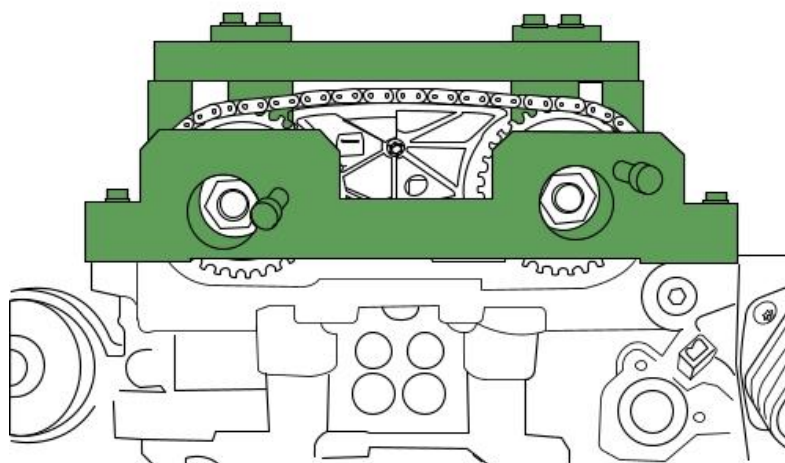
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Inlet camshaft:



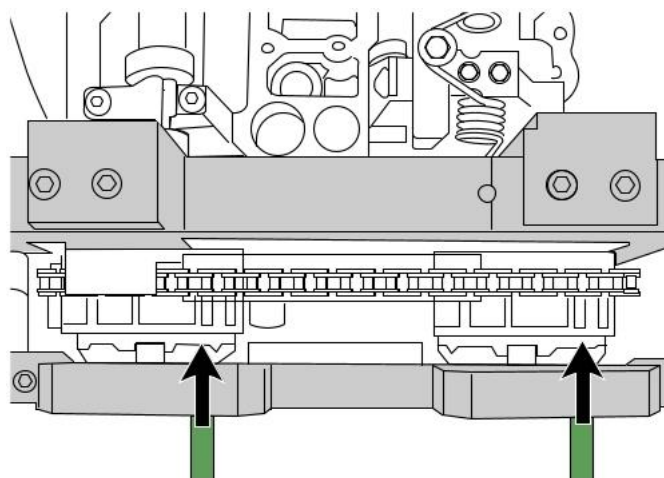
Fit the special tool (2 212 830)



Loosen the camshaft gearwheel bolts

Remove the chain tensioner

Turn the camshaft gearwheels until the locking tool fits

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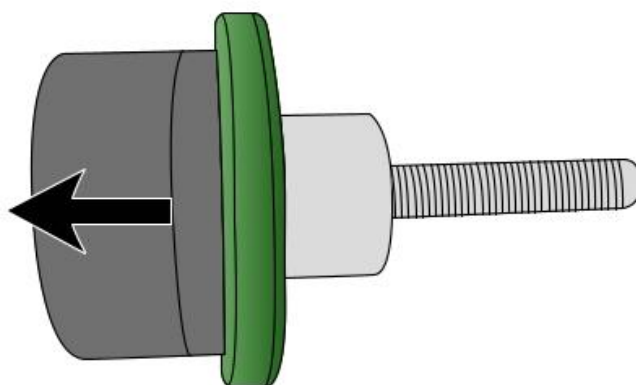
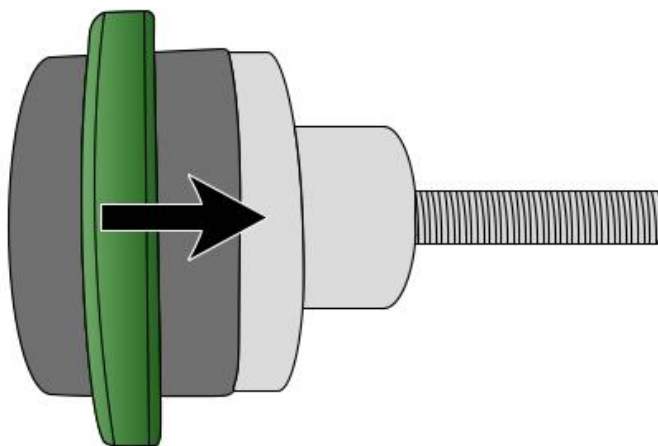
Fit the dummy chain tensioner (11 9 340)

Pre-tension the timing chain (0.6 Nm)

Tighten the camshaft gearwheel bolts

Remove the special tools (11 9 340, 2 219 548, 2 212 831, 2 212 830)

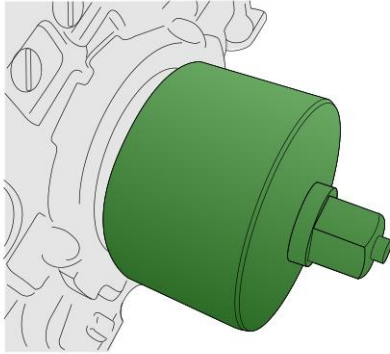
Fit the crankshaft seal over the special tool (2 212 822)



Secure the special tool with two bolts (2 212 822)

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Assemble the special tool as shown. Tighten the special tool bolt until full surface contact is made.



Further refitting is in the reverse order of removal

Refill the engine oil

Engine sump, including filter 5.0 (l)

Read out the fault codes

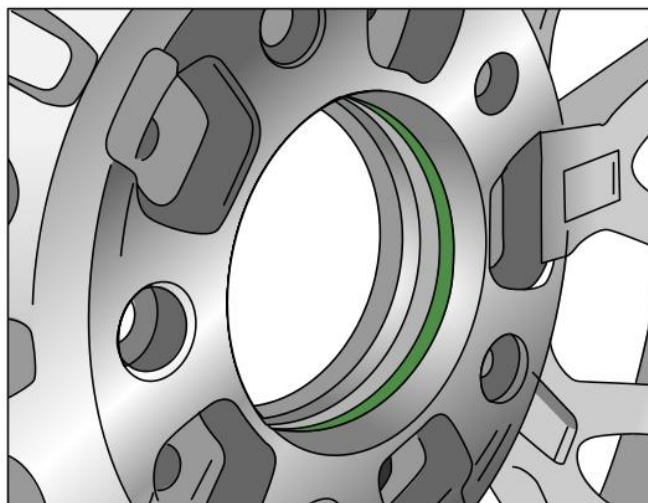
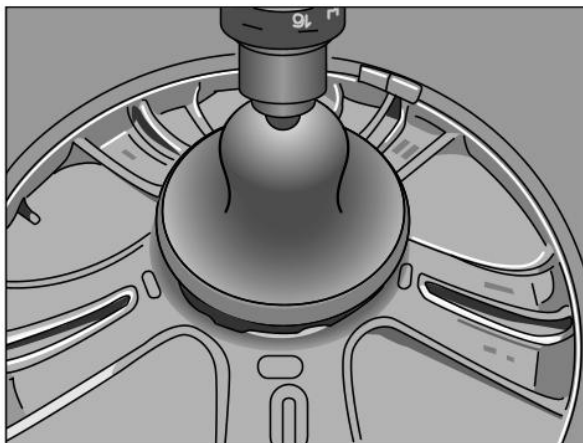
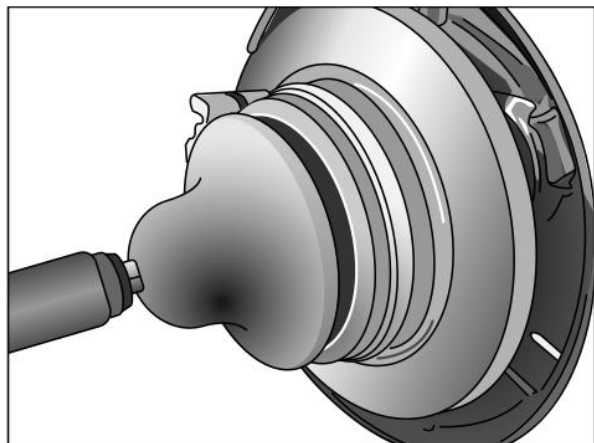
Torque settings

Wheel bolts

Wheel bolts (Clean the contact surfaces and wheel bolts/nuts; lightly grease the wheel centre taper only but do not grease the wheel bolts/nuts)

Wheel bolts (Caution! Do not use power tools when screwing in or tightening wheel nuts/bolts)

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Wheel bolts 140 (Nm)

Timing chain tensioner

Timing chain tensioner (Upper) 80 (Nm)

Renew the seal (Threads lightly oiled)

Timing chain guide

Timing chain guide (M6) 10 (Nm)

Timing chain guide (M7) 14 (Nm)

Timing chain guide (M8) 24 (Nm)

Camshaft gearwheel

Camshaft gearwheel

Stage 1 55 (Nm)

Stage 2 55 (°)

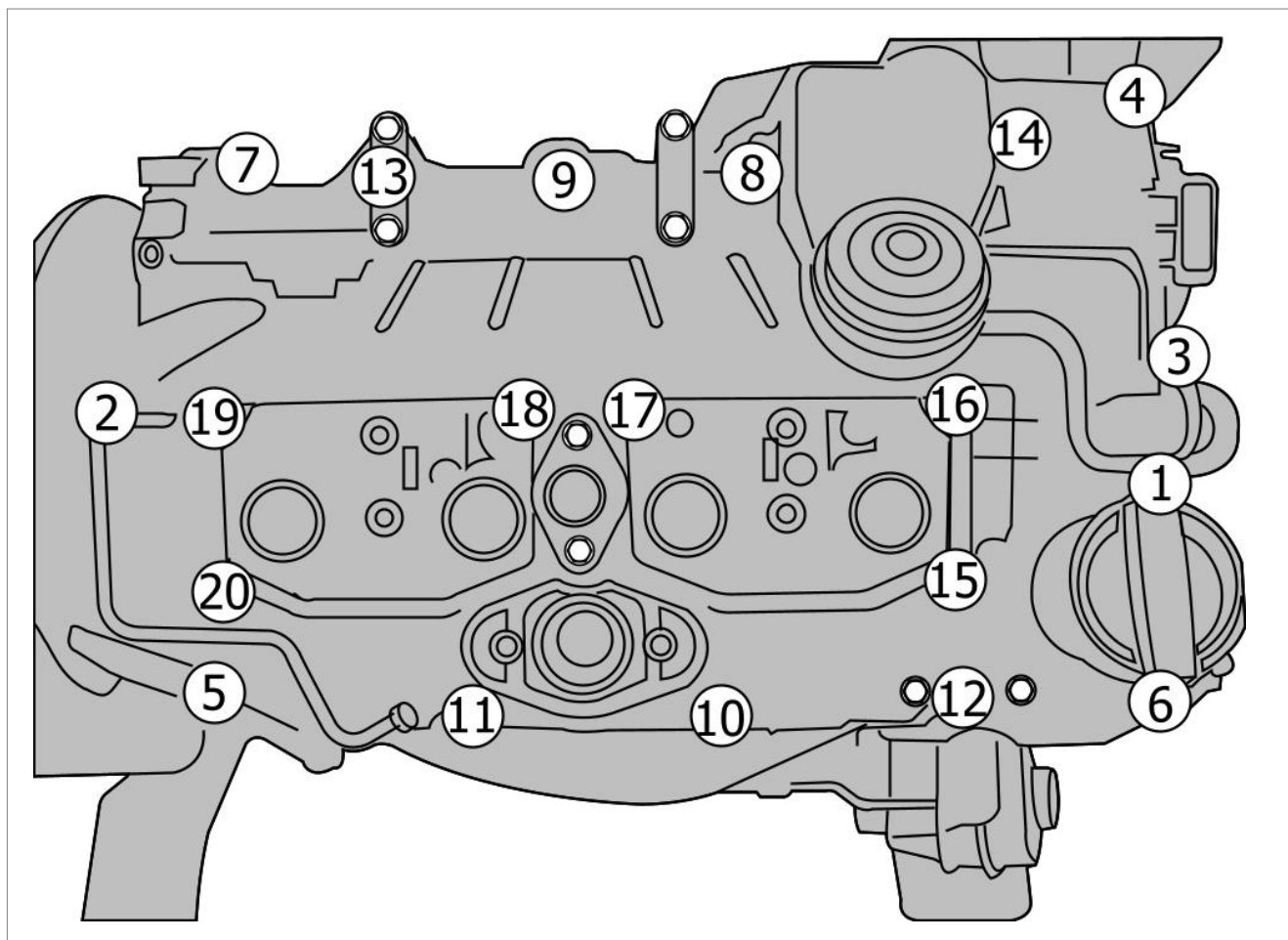
Crankshaft pulley

Crankshaft pulley (Outer bolts) 35 (Nm)

Valve cover

Valve cover 9.2 (Nm)

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Camshaft position sensor

Camshaft position sensor 8 (Nm)

Oil sump

Sump 22 (Nm)

Spark plugs

Spark plugs (M12) 20 - 26 (Nm)

Spark plugs (M14) 27 - 33 (Nm)

Fuel rail

Fuel rail 6 - 8 (Nm)

High-pressure pump

High-pressure pump 12 (Nm)

Ignition coil

Ignition coil 10 (Nm)

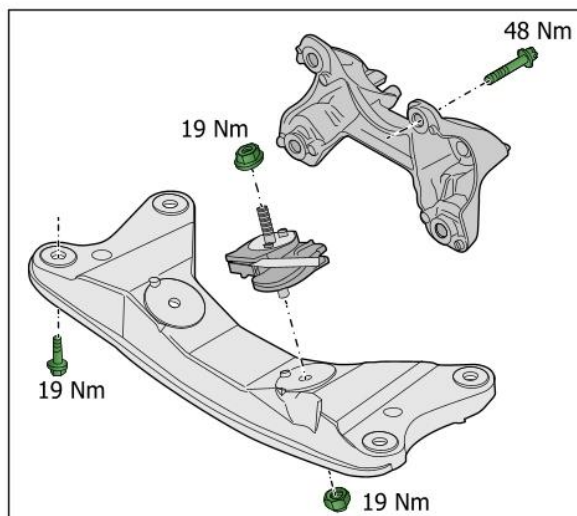
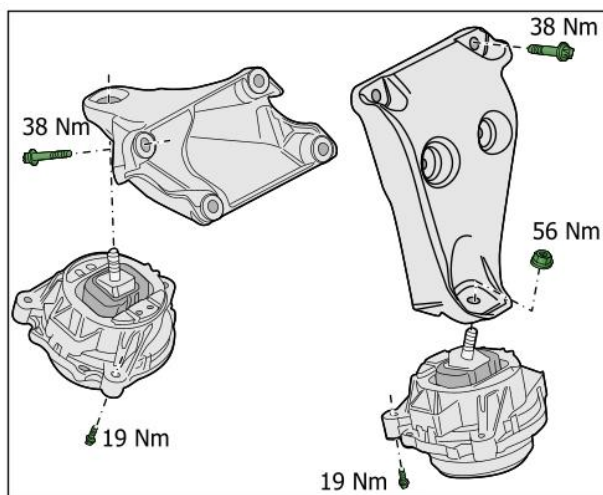
Front drive shaft to wheel bearing housing

Suspension: Refer to 'Technical drawings'

Steering: Refer to 'Technical drawings'

Engine/gearbox mounts

Engine

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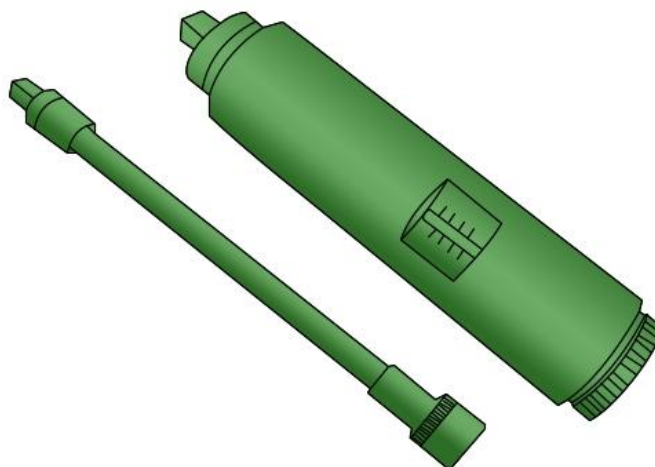
Vacuum pump: (10 Nm)

High-pressure pipe: (M12: 20 - 26 Nm)

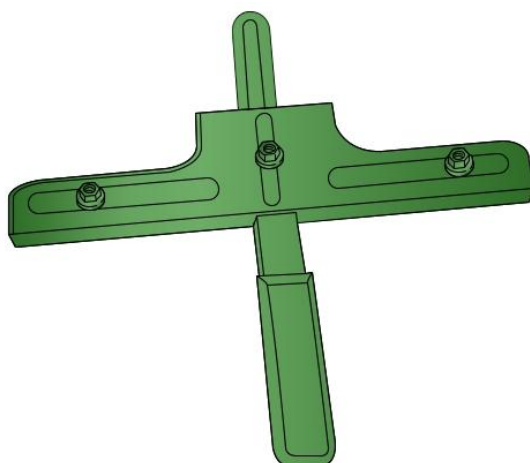
High-pressure pipe: (M14: 27 - 33 Nm)

Special tools

Adjustment tool: (00 9 250)

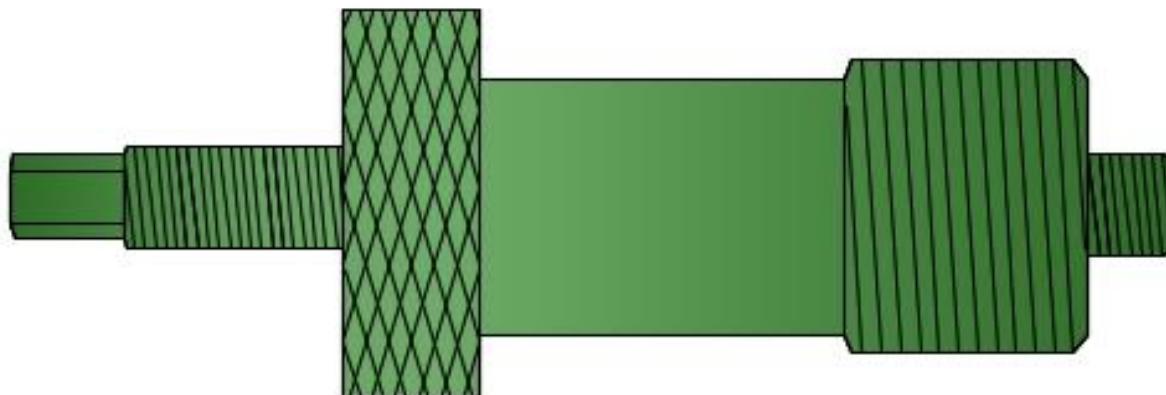


Crankshaft locking tool: (11 8 180)

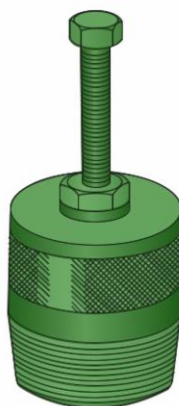


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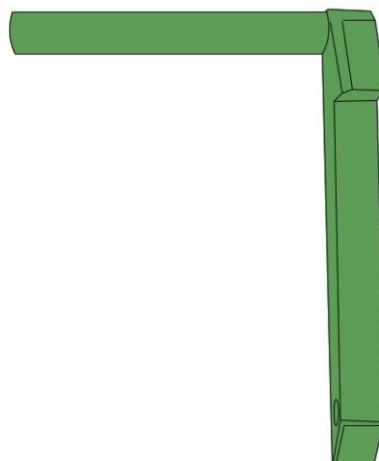
Adaptor: (11 9 340)



Extractor: (11 0 371, 11 0 372)

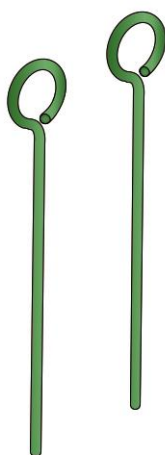


Flywheel locking tool: (2 219 548)

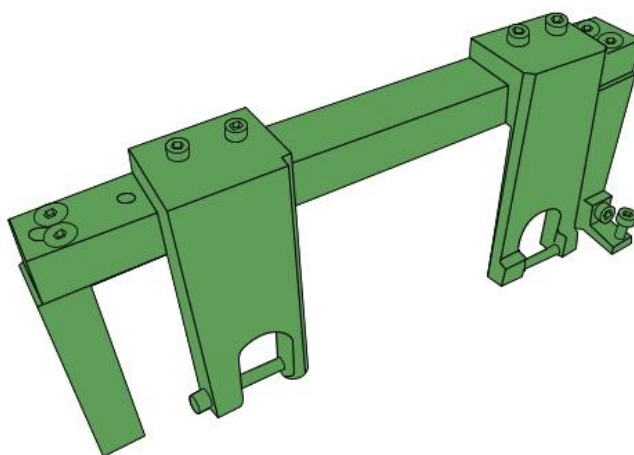


Locking pin: (11 4 120)

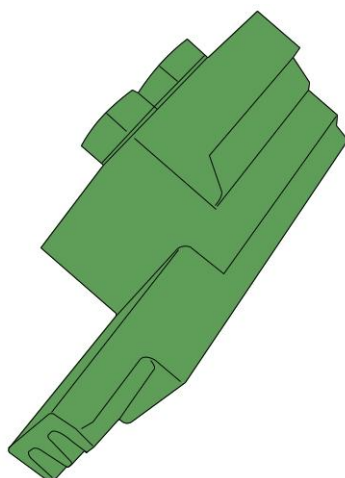
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Camshaft timing tool: (2 212 831)

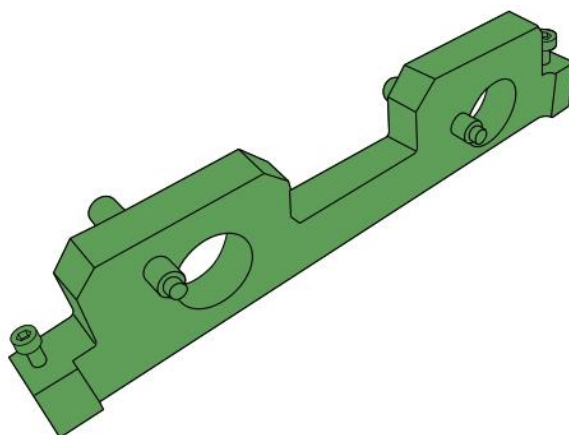


Crankshaft locking tool: (22 2 742)

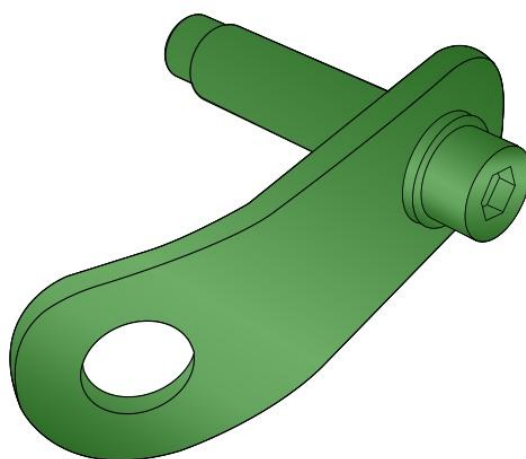


Camshaft timing tool: (2 212 830)

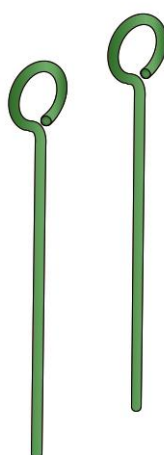
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Special tool: (2 212 825)

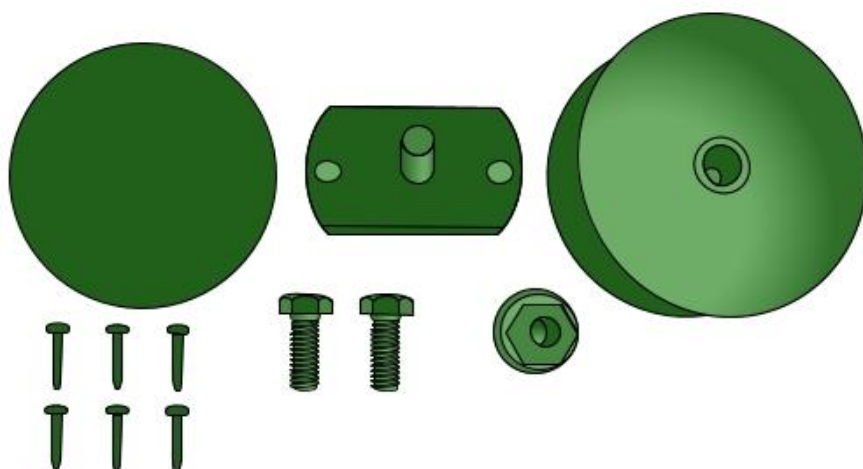


Chain tensioner locking pin: (11 3 340)

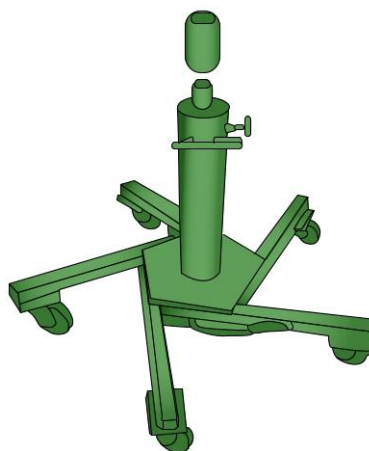


Crankshaft front radial seal fitting tool: (11 9 231, 11 9 232, 11 9 233, 11 9 234)

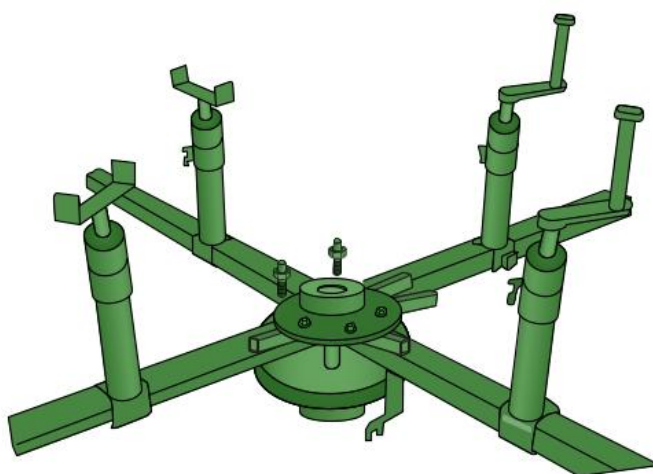
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Transmission jack: (00 2 030)



Subframe support: (31 5 250)



Crankshaft front radial seal fitting tool: (2 212 822)

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